

FEMINA K

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building AI-driven systems, full-stack applications, and intelligent automation platforms. Developed production-ready applications, RAG-based knowledge retrieval systems, and AI-integrated workflows deployed on modern cloud infrastructure.

Experienced in designing scalable backend architectures, REST APIs, secure authentication systems, and database models. Skilled in integrating large language models, vector databases, embedded systems, and event-driven automation to solve real-world challenges efficiently. Focused on building reliable, practical, and impactful technology from concept to deployment.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, SQL, C/C++, Embedded C

Frontend: React, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, Flask, Django, FastAPI

Databases: MySQL, PostgreSQL, MongoDB, Supabase

Embedded & IoT: Arduino UNO, Microcontroller Interfacing, Peripheral Integration, Wokwi Simulation

Cloud: Google Cloud Console, App Engine, Railway, Render, Heroku, Vercel, Netlify

Automation: n8n, Zapier, Make.com, Selenium

AI APIs: OpenAI, Gemini, Claude, Deepseek, Kimi, Perplexity, Ollama

Integrations: Google APIs, Stripe, Razorpay, Gmail API, Twilio API, Slack API

Security: JWT, OAuth 2.0, RBAC

Tools: Git, GitHub, Pandas

PROJECTS

ActiveTime Production-Ready Productivity Platform

- Engineered complete MERN stack architecture for scalable task and productivity management.
- Designed JWT-secured authentication and protected API routes.
- Structured MongoDB schemas for efficient data handling.
- Deployed frontend on Vercel and backend on Google Cloud with secure configuration.

Aletheia AI-Powered Knowledge Retrieval Platform

- Implemented semantic vector search using Qdrant for RAG-based retrieval.
- Built secure multi-tenant authentication with JWT and RBAC.
- Designed scalable document ingestion pipeline with Celery and Redis.
- Secured 2nd Place in hackathon for AI-driven retrieval innovation.

Automated TradingView Idea Posting System

- Designed end-to-end AI automation workflow using n8n and Gemini API.
- Implemented structured JSON extraction via advanced prompt engineering.
- Built browser automation layer using Playwright.
- Engineered validation and duplicate-prevention logic for reliability.

Smart Password-Based Access Control System (IoT Simulation)

- Developed embedded password validation logic in Embedded C.
- Integrated keypad, LCD, servo motor, and buzzer modules.
- Simulated full hardware architecture using Wokwi.
- Structured modular embedded code for maintainability and scalability.

Speech Segmentation Tool Audio Processing System

- Implemented silence-based speech detection using FFmpeg.
- Automated audio segmentation into structured .wav outputs.
- Generated JSON metadata with timestamps for processing pipelines.
- Refactored into modular production-ready Python tool with validation handling.

EDUCATION

B.E. Computer Science & Engineering
Thanthai Periyar Government Institute of Technology, Vellore
– Relevant Coursework: Data Structures, DBMS, Operating Systems, Computer Networks

2023 2027
CGPA: 8.5

ACHIEVEMENTS

- Winner MINDMAZE, Ananta Bodha 2K24
- 2nd Place Innovators Arena 2026
- Neurathon 26 Finalist Top 200 Teams
- HackNight 25 Participant 36-Hour Hackathon
- National Symposium XPLORE 24

AREAS OF INTEREST

Artificial Intelligence Full-Stack Development Agentic Workflows Scalable Backend Systems Cloud Deployment
Human-Centered Product Design

SOFT SKILLS

Systems Thinking Analytical Debugging Clear Technical Communication Ownership Mindset Collaborative
Teamwork Adaptability Time Management